.1 Semi-synthetic experiment with NHANES patients

Data and simulator. We build a semi-synthetic benchmark in which the patients are real and only their responses to insulin are simulated. We take the 229 insulin-naive adults with diagnosed diabetes and HbA1c $\geq 7\%$ in the National Health and Nutrition Examination Survey (NHANES) 2017–2018 [Centers for Disease Control and Prevention, National Center for Health Statistics, 2020], i.e., the patients for whom basal insulin initiation is typically considered (Table 1). Each simulated patient is drawn with replacement from this cohort and keeps their measured baseline HbA1c A_0 and body weight W. For a daily basal insulin dose $d_k \in \{0, 10, 20, 30, 40\}$ U/day ($K = 4$), the HbA1c after treatment follows an $E_{\max}$ model on the excess HbA1c above 5%,

$$Y_k = 5 + (A_0 - 5)\Big(1 - E_{\max}\frac{S\,d_k}{ED_{50} + S\,d_k}\Big) + \varepsilon, \qquad S = \frac{\bar{W}}{W}\,e^{\eta},$$

with $E_{\max} = 0.7$, $ED_{50} = 10$ U/day, $\bar{W} = 86.2$ kg the median cohort weight, so that the effect depends on the dose per kilogram, $\eta \sim N(0, 0.3^2)$ unexplained between-patient variation, and $\varepsilon \sim N(0, 0.15^2)$ assay noise shared across doses. With these values, the mean HbA1c reduction is 1.2, 1.5, 1.7, 1.9 percentage points at 10, 20, 30, 40 U/day, respectively (0.12–0.46 U/kg for a patient of median weight), of the same order as the reductions reported when basal insulin is added to oral therapy [Riddle et al., 2003]. The parameters of the dose response are chosen to be clinically plausible but are not fitted to trial data. We use the ADA target HbA1c $\leq 7\%$ [American Diabetes Association Professional Practice Committee, 2025] as the benefit ($y^u = 7.0$) and the intensive target HbA1c $\leq 6\%$, which increased mortality and severe hypoglycemia in the ACCORD trial [Action to Control Cardiovascular Risk in Diabetes Study Group, 2008], as the harm ($y^l = 6.0$). Since Y_k is decreasing in k for every patient, Assumptions **??** and **??** hold. Assumption **??** is not guaranteed because the latent variable $(A_0, W, \eta, \varepsilon)$ is multi-dimensional; because the simulator returns all potential outcomes, we compute the true θ^{MNH} from the joint counterfactual outcomes and compare it with the identified expression in Theorem **??**, finding a maximum discrepancy of 1.3×10^{-3}.

Table 2 shows the ground truth. The Benefit criterion is maximized at 40 U/day and the Harmless criterion at 20 U/day, whereas the MND-B is 10 U/day and the MND-H is 10 U/day: 10 U/day is exactly necessary for the largest share (51.1%) of patients. At 40 U/day, 21.5% of patients are driven to HbA1c $\leq 6\%$ and 77.9% are over-treated, compared with 1.0% and 5.5% at 10 U/day.

Results. We set $T^{\mathrm{initial}} = 10$ and $T = 10000$, and repeat each experiment 100 times. Table 3 shows the chosen doses, Figures 1 and 2 the cumulative regret, and Table 4 the outcomes of the trial patients. MND-B UCB identifies the MND-B in 39%, 73% and 100% of runs with $\kappa = 1$, 0.1 and 0.01, respectively. Benefit UCB converges to 40 U/day and Harmless UCB to 20 U/day, and the doses chosen by the MED-based methods depend on ϕ. Compared with Benefit UCB, MND-B UCB ($\kappa = 0.01$) administers 9.8 instead of 36.0 U/day on average and reduces over-intensive control (HbA1c $\leq 6\%$) from 18.7% to 1.4% and over-treatment from 73.6% to 7.2%, while 53.1% instead of 80.1% of patients reach the target. This reflects the trade-off that the MND makes explicit: it targets the dose that is necessary for the largest share of patients rather than the dose that brings the most patients to target.

MND-H UCB identifies the MND-H in 0%, 0% and 0% of runs with $\kappa = 1$, 0.1 and 0.01, and in 0% with $\kappa = 0.003$ and 100% with $\kappa = 0.001$. The difficulty is the confidence radius of dose 0, $r_0 = \sum_{k=1}^{K} r_k$. When dose 0 leads, its confirmation partners are sampled only until their counts reach about $\log(t)/\kappa$ (Step 1 of the proof of Theorem **??**), so each r_k stays near $2\sqrt{\kappa}$ and r_0 is of order $2K\sqrt{\kappa}$; dose 0 is therefore abandoned only if roughly $\kappa \lesssim (\Delta_0/2K)^2 \approx 0.0012$ in this setting ($\Delta_0 = 0.278$). For MND-B UCB, the radius of dose 0 involves only doses 0 and K and stays near $\sqrt{2\kappa}$, giving the much weaker requirement $\kappa \lesssim \Delta_0^2/2 \approx 0.04$. With κ below this threshold, MND-H UCB identifies the MND-H, but a smaller κ also lengthens confirmation (the term $\log(T)/\kappa$ in Theorem **??**), so in this setting MND-H UCB requires κ in a narrow range that depends on the unknown gaps. This limitation grows with K, consistent with the factor K^2 in Theorem **??**; a tighter confidence bound for $\theta^{\mathrm{MNH}}(0)$ that exploits the telescoping structure of $\sum_k \theta^{\mathrm{MNH}}(k)$ is an interesting direction for future work.

References

Action to Control Cardiovascular Risk in Diabetes Study Group. Effects of intensive glucose lowering in

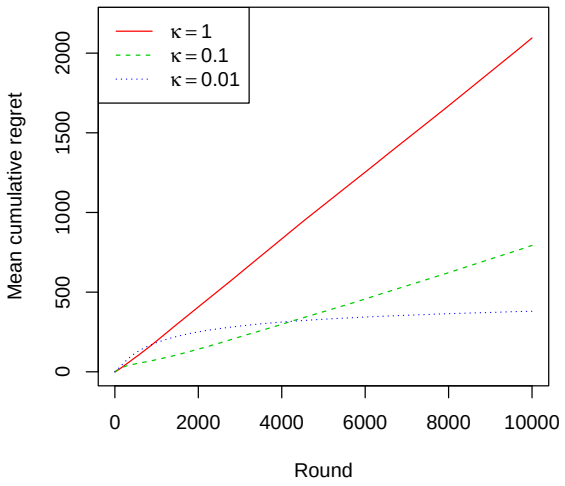


Figure 1: Mean cumulative regret of MND-B UCB in the semi-synthetic NHANES experiment over 100 runs (red solid: $\kappa = 1$; green dashed: $\kappa = 0.1$; blue dotted: $\kappa = 0.01$).

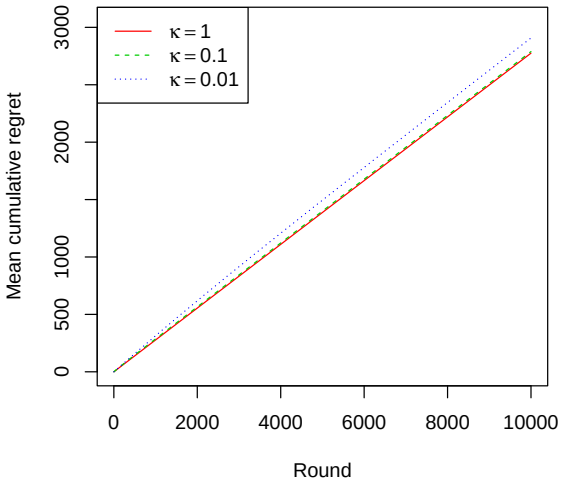


Figure 2: Mean cumulative regret of MND-H UCB in the semi-synthetic NHANES experiment over 100 runs (red solid: $\kappa = 1$; green dashed: $\kappa = 0.1$; blue dotted: $\kappa = 0.01$).

Table 1: Real cohort used in the semi-synthetic experiment: insulin-naive adults with diagnosed diabetes and HbA1c $\geq 7\%$ in NHANES 2017–2018 ($n = 229$). Median [interquartile range].

Age (years)	62.0 [53.0, 69.0]
Body weight (kg)	86.2 [73.3, 102.4]
Baseline HbA1c (%)	7.8 [7.3, 8.9]

type 2 diabetes. *New England Journal of Medicine*, 358(24):2545–2559, 2008.

American Diabetes Association Professional Practice Committee. 6. glycemic goals and hypoglycemia: Standards of care in diabetes–2025. *Diabetes Care*, 48(Supplement 1), 2025.

Centers for Disease Control and Prevention, National Center for Health Statistics. National health and nutrition examination survey data, 2017–2018. Hyattsville, MD: U.S. Department of Health and Human Services, 2020. Files DEMO_J, DIQ_J, GHB_J and BMX_J.

Matthew C. Riddle, Julio Rosenstock, and John Gerich. The treat-to-target trial: Randomized addition of glargine or human NPH insulin to oral therapy of type 2 diabetic patients. *Diabetes Care*, 26(11): 3080–3086, 2003.

Table 2: Ground truth of each basal insulin dose in the semi-synthetic experiment, computed from 2×10^6 patients resampled from the NHANES cohort. Over-treated: the patient would also have reached the target at the next lower dose. Bold: the optimum of each criterion.

Dose (U/day)	0	10	20	30	40
θ^{B} (HbA1c $\leq 7\%$)	0.055	0.566	0.709	0.779	**0.823**
θ^{H} ($6\% <$ HbA1c $\leq 7\%$)	0.055	0.556	**0.638**	0.632	0.609
θ^{MNB}	0.231	**0.511**	0.143	0.070	0.044
θ^{MNH}	0.233	**0.510**	0.143	0.070	0.044
HbA1c $\leq 6\%$	0.000	0.010	0.071	0.147	0.215
Over-treated	0.000	0.055	0.566	0.709	0.779

Table 3: Most frequently chosen dose in the semi-synthetic experiment ($T = 10000$, $T^{\mathrm{initial}} = 10$, 100 runs) and the proportion of runs in which it is chosen. The MND-B is 10 U/day and the MND-H is 10 U/day.

Algorithm (Benefit)	Chosen dose	Algorithm (Harmless)	Chosen dose
MND-B UCB ($\kappa = 1$)	**10 U/day** (39%)	MND-H UCB ($\kappa = 1$)	0 U/day (100%)
MND-B UCB ($\kappa = 0.1$)	**10 U/day** (73%)	MND-H UCB ($\kappa = 0.1$)	0 U/day (100%)
MND-B UCB ($\kappa = 0.01$)	**10 U/day** (100%)	MND-H UCB ($\kappa = 0.01$)	0 U/day (100%)
MND-B UCB without confirmation task	**10 U/day** (43%)	MND-H UCB without confirmation task	0 U/day (100%)
Benefit UCB	40 U/day (100%)	Harmless UCB	20 U/day (65%)
MED-TS for benefit ($\phi = 0.5$)	**10 U/day** (99%)	MED-TS for harmless ($\phi = 0.5$)	**10 U/day** (100%)
MED-TS for benefit ($\phi = 0.7$)	20 U/day (72%)	MED-TS for harmless ($\phi = 0.7$)	20 U/day (70%)
MED-TS for benefit ($\phi = 0.8$)	40 U/day (100%)	MED-TS for harmless ($\phi = 0.8$)	20 U/day (79%)
TS_A for benefit ($\phi = 0.5$)	**10 U/day** (100%)	TS_A for harmless ($\phi = 0.5$)	**10 U/day** (100%)
TS_A for benefit ($\phi = 0.7$)	20 U/day (100%)	TS_A for harmless ($\phi = 0.7$)	20 U/day (70%)
TS_A for benefit ($\phi = 0.8$)	30 U/day (57%)	TS_A for harmless ($\phi = 0.8$)	20 U/day (69%)

Table 4: In-trial outcomes over the $T = 10000$ trial patients, averaged over 100 runs: mean basal insulin dose, and the proportions of patients who reached the target (HbA1c $\leq 7\%$), were driven to over-intensive control (HbA1c $\leq 6\%$), and were over-treated (would also have reached the target at the next lower dose).

Algorithm	Mean dose (U/day)	HbA1c $\leq 7\%$	HbA1c $\leq 6\%$	Over-treated
MND-B UCB ($\kappa = 0.01$)	9.8	53.1%	1.4%	7.2%
MND-H UCB ($\kappa = 0.01$)	8.1	27.1%	3.6%	17.1%
MND-H UCB ($\kappa = 0.003$)	18.2	54.0%	8.1%	38.4%
MND-H UCB ($\kappa = 0.001$)	23.1	66.8%	10.2%	48.6%
Benefit UCB	36.0	80.1%	18.7%	73.6%
Harmless UCB	26.3	74.1%	11.9%	61.4%
MED-TS for benefit ($\phi = 0.7$)	23.4	73.3%	9.7%	61.1%
TS_A for benefit ($\phi = 0.7$)	20.4	71.0%	7.4%	56.7%